Journal of Molecular Medicine
https://doi.org/10.1007/s00109-020-01935-6

JMolMed

REVIEW

# Lysosomal storage diseases: current therapies and future alternatives

Andrés Felipe Leal[1] · Angela Johana Espejo-Mojica[1] · Oscar F. Sánchez[2] · Carlos Manuel Ramírez[3] · Luis Humberto Reyes[3] · Juan C. Cruz[4] · Carlos Javier Alméciga-Díaz[1]

Received: 7 January 2020 / Revised: 28 May 2020 / Accepted: 3 June 2020


## Abstract

Lysosomal storage disorders (LSDs) are a group of monogenic diseases characterized by progressive accumulation of undegraded substrates into the lysosome, due to mutations in genes that encode for proteins involved in normal lysosomal function. In recent years, several approaches have been explored to find effective and successful therapies, including enzyme replacement therapy, substrate reduction therapy, pharmacological chaperones, hematopoietic stem cell transplantation, and gene therapy. In the case of gene therapy, genome editing technologies have opened new horizons to accelerate the development of novel treatment alternatives for LSD patients. In this review, we discuss the current therapies for this group of disorders and present a detailed description of major genome editing technologies, as well as the most recent advances in the treatment of LSDs. We will further highlight the challenges and current bioethical debates of genome editing.

**Keywords** Lysosomal storage disorders · Enzyme replacement therapy · Genome editing · CRISPR/Cas9

## Introduction

Lysosomal storage diseases (LSDs) are a group of about 50 inborn errors of metabolism caused by mutations in genes encoding for proteins necessary for lysosomal function, such as hydrolases, integral membrane proteins, cofactors, transferases, transporters, and enzymes activators, as well as several non-lysosomal proteins that participate in lysosome-associated processes [1]. Function loss of lysosomal proteins leads to the dysregulation of several cellular processes linked to this organelle, which includes membrane repair, autophagy, exocytosis, lipid homeostasis, signaling cascades, or cell viability [2]. These disorders are usually classified depending on the primary undegraded substrate that is accumulated, e.g., sphingolipids, glycosaminoglycans, and glycogen; LSDs can also be classified as a function of the nature of the underlying molecular defect, e.g., single hydrolytic enzyme deficiency, protease enzyme deficiency, cofactor/activator deficiency, and multiple enzyme deficiencies [2, 3]. Combined birth prevalence of LSDs can range from 7.5 per 100,000 to 23.5 per 100,000 people [4].

Support therapy is the typical strategy for LSD patients' care, including ventilatory support; surgical interventions; nutritional support; physical therapy; and managing of neurological, cardiac, and renal complications. Besides, specific treatments have been developed for some LSDs to correct the metabolic defect, including enzyme replacement therapy (ERT), hematopoietic stem cell transplantation (HSCT), pharmacological chaperones (PC), substrate reduction therapy (SRT), and gene therapy [3, 5–9]. In this review, we discuss the most relevant therapies for LSDs (Fig. 1), focus on new treatment alternatives, and giving special attention to genome editing strategies, including their challenges and bioethical implications. For this purpose, the PubMed, Embase, and SpringerLink databases were consulted, limiting the search to articles written in English and ranged between 2010 and 2020. The search was conducted using the following keywords and Booleans operators: ("lysosomal storage diseases" AND "therapeutics") OR ("lysosomal storage disease" AND

✉ Carlos Javier Alméciga-Díaz
cjalmeciga@javeriana.edu.co

1 Institute for the Study of Inborn Errors of Metabolism, Faculty of Science, Pontificia Universidad Javeriana, Cra. 7 No. 43-82 Building 54, Room 305A, Bogotá D.C 110231, Colombia

2 Neurobiochemistry and Systems Physiology, Biochemistry and Nutrition Department, Faculty of Science, Pontificia Universidad Javeriana, Bogotá D.C., Colombia

3 Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá D.C., Colombia

4 Department of Biomedical Engineering, Universidad de los Andes, Bogotá D.C., Colombia

Published online: 11 June 2020

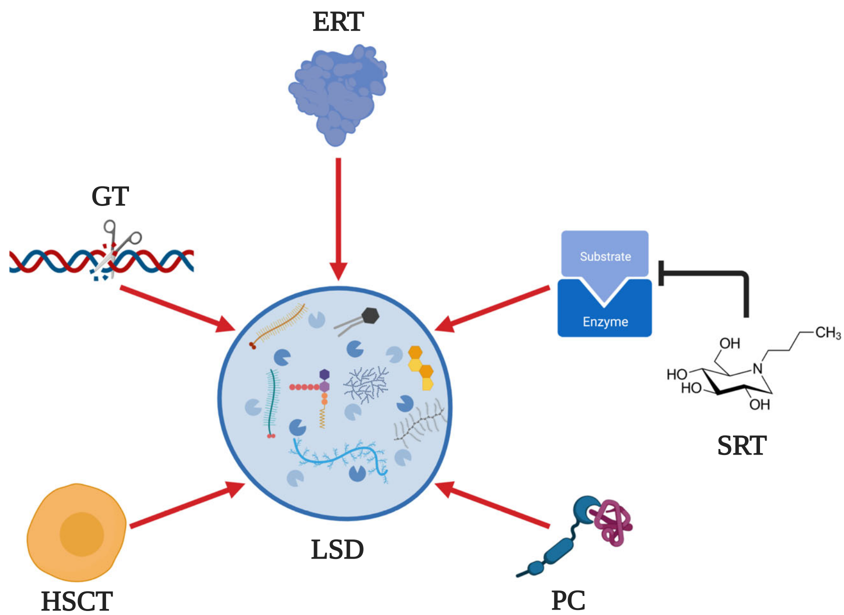


**Fig. 1** Therapeutic alternatives for LSD. At the date, several strategies have been developed to reduce the pathophysiological effect of substrate accumulation into lysosome (i.e., glycosaminoglycans, glycolipids, glycogen, and cholesterol). Enzyme replacement therapy (ERT) consists of the administration of lysosomal enzymes that are deficient in patients with LSD. Substrate reduction therapy (SRT) implies the inhibition of an enzyme responsible for the synthesis of the stored substrate. Since most of the mutations lead to protein misfolding, pharmacological chaperones (PCs) have also been explored for LSDs treatment. Hematopoietic stem cell transplantation (HSCT) and gene therapy (GT) have been postulated as promising therapies through which it could be possible to cure the LSD. This figure was created with BioRender.com

"genome editing") OR ("lysosomal storage disease" AND "CRISPR") OR ("lysosomal storage disease" AND "enzyme replacement therapy") OR ("Bioethics" AND "Genome editing").

## Enzyme replacement therapy

After lysosome discovery by de Duve in 1955 [10], the identification of the cross-correction mechanism [11] and protein internalization via mannose and mannose-6-phosphate (M6P) receptors [5] paved the way for the development of ERT and other treatment alternatives for LSDs. Early clinical trials of ERT during the 1990s for Gaucher patients [5] were based on the correction of the metabolic defect by the administration of a purified enzyme from human placenta [5]. Since then, different efforts to optimize the ERT for Gaucher's diseases have demonstrated their effectiveness and safety [12], which has been crucial to delineate the ERT basis for other LSDs. Consequently, ERT, enabled by recombinant DNA technology, has been approved for Gaucher, Fabry, and Pompe diseases, late infantile neuronal ceroid lipofuscinosis type II, acid lipase deficiency, alpha-mannosidosis, and mucopolysaccharidoses (MPS) type I, II, IVA, VI, and VII [3, 5]. Given that ERT is intravenously administered, the main drawback of this therapy is its limited ability to cross the blood-brain barrier (BBB), which has prevented the improvement of neurologic disorders associated with some LSDs [5].

One critical aspect for the ERT success is to assure that the therapy can reach the affected tissue. For instance, in Fabry disease, ERT easily reaches accessible vascular endothelium, cardiomyocytes, and renal podocytes[13]. Conversely, in MPS, the ERT is required to penetrate the connective tissue matrix and cells (i.e., chondrocytes) or reach the central nervous system (CNS), which turns out to be refractive tissues to the translocation of most molecules [5]. Accumulating evidence has shown that ERT efficacy lies in the presence of mannose-phosphate and M6P receptors in the target cell and the generation of neutralizing anti-ERT antibodies, thereby causing significant dose variations among patients and LSDs to keep the minimum effective therapeutic level [14]. ERT is an appealing strategy for those patients that retain some enzyme activity due to their reduced immunological response. For instance, in Gaucher disease type 1, most patients keep some of the β-glucocerebrosidase activity. As a result, only 15% of patients raised antibodies against the ERT [15]. Conversely, in patients with Fabry disease, about 30–50% of them present a severe immune response to the therapy [13, 16], while in MPS up to 100% of the patients may develop anti-ERT antibodies [8].

**Proinflammatory events in LSD and ERT** Several proinflammatory pathways have been described in LSDs as a secondary consequence of the lysosome storage [17]. For instance, microglial activation along with Fc receptor γ-dependent astrocytosis have been observed in GM2 gangliosidosis [18], whereas in MPS IIIB, partially digested GAGs are sensed by the cells as a danger-associated molecular pattern in the context of the Toll-like receptor 4 with an increment of proinflammatory cytokines like TNF-α, IL-1β, and MIP-1α [19–21]. In addition to these intrinsic inflammatory effects caused by the disease, recent evidence suggests that the use of anti-inflammatory compounds may be necessary to obtain a successful outcome during ERT. This has been attributed to secondary inflammatory processes induced by the accumulated substrate *per se* and the infusion of the therapeutic enzyme [22, 23]. In this regard, the use of antagonists of the TLR4/TNF-α pathway has improved clinical features and the ERT therapeutic effect in preclinical and clinical trials of MPS I and VI [24–26], thereby suggesting that a comprehensive approach might be helpful.

## Substrate reduction therapy

Substrate reduction therapy (SRT) uses small molecules to inhibit the synthesis of substrates to reduce the storage of the involved metabolites [27, 28]. SRT has been mainly exploited in the treatment of sphingolipidoses; however, some of the developed approaches have been evaluated for MPS. For instance, genistein, genistein-derived compounds, rhodamine B, and small interference RNA have been assessed for MPS [8]. Meanwhile, several small molecules (e.g., N-butyldeoxynojirimycin, miglustat, and eliglustat) able to inhibit the glucosylceramide synthase, which is involved in the initial synthesis of glycosphingolipids, have been identified for the treatment of glycolipidoses [27, 29]. This approach's potential benefits include oral delivery, BBB crossing, and a pharmacodynamic response that is generally complementary to ERTs. SRT, as described for ERT, works best in patients with residual enzymatic activity.

**SRT for the treatment of sphingolipidoses** SRT efficacy has been confirmed in a murine model of Gaucher type I, in which combined use of SRT and ERT led to exceedingly low levels of the accumulated substrates [30, 31]. Likewise, Gaucher type I patients, under a single or double SRT dose regime per day, have shown a reduction in the hepatosplenomegaly along with an increment in hematological parameters and bone mineral density [32–35]. SRT has been explored in Gaucher type III patients with minimal improvement in pulmonary condition [36]. In Fabry disease, a strong SRT dose-response was detected in the reduction of the accumulated substrate, but with substantial untoward neurological toxicity [37]. On the other hand, SRT showed modest or non-neural improvement in Niemann-Pick disease type C and infantile and juvenile GM2 gangliosidosis [38, 39]. Alternatively, an SRT approach based on cyclodextrins was developed to treat Niemann-Pick disease type C1. In this case, cyclodextrins reduced the intracellular accumulation of cholesterol, but as opposed to conventional SRTs, these compounds failed to inhibit the biosynthesis of the accumulating substrate [40, 41]. SRT holds a grand promise for the treatment of LSDs with neurological impairments. Nevertheless, impactful advances will only come to light if new molecules are able to cross the BBB without affecting the brain lipid balance.

**Substrate degradation enzyme therapy as an alternative to SRT** New therapeutic options using recombinant enzymes have been described to reduce substrate accumulation by its degradation. In this context, a novel therapy called substrate degradation enzyme therapy (SDET) has been recently introduced, where a thermostable keratanase (TSK) isolated from *Bacillus circulans* administered to an MPS IVA murine model for a reduction of ~ 70% of mono-sulfated KS in serum. Moreover, SDET reduced vacuolization of chondrocytes from 67 to 29%, thereby suggesting that this approach could overcome the limited efficacy of ERTs in the beneficial *in vivo* breakdown of KS on cartilage [42]. The identification of novel enzymes with the potential to degrade the accumulated substrates in different LSDs will thus provide alternative treatments LSDs for which ERT or SRT is not possible.

## Pharmacological chaperone therapy

Single-nucleotide polymorphisms are generally responsible for altering the tertiary and quaternary structure of proteins, which in turn leads to protein destabilization and degradation by the endoplasmic reticulum-associated degradation and ultimately the proteasome [43]. In this sense, PCs are small molecules that bind to a mutant protein and promote its correct folding and intracellular trafficking [44]. These molecules bind to the target protein, usually the active cavity, in the endoplasmic reticulum, where they act as molecular templates to preserve the native conformation of the protein or promote the proper folding and trafficking of a mutant protein [44]. PCs have advantages over ERT such feasibility for oral administration, wide tissue distribution profile, and fewer immunogenicity issues [45]. In LSDs, PCs have been assessed in Fabry disease [46], Gaucher disease, Pompe disease, gangliosidosis (GM1 and GM2)[47, 48], and MPS type II, IIIC, IVA, and IVB [49–51]. Recent studies have proposed that PCs work better when involved mutations are harmless to (i) resides involved in the ligand, substrate, or cofactor binding; (ii) cysteines involved in disulfide bridges; (iii) prolines involved in the formation or elimination of turns; (iv) formation of hydrogen bonds; and (v) the size of amino acid since substitutions by bulkier residues are not appealing mainly due to restricted conformational freedom [48]. With this set of recommendations, the role of possible mutations on the conformation and integrity of active sites could be evaluated with the aid of molecular modeling tools, to select PC-responsive mutations.

**Discovery and the emergence of PCs** PCs were first evaluated in Fabry disease, in which the use of α-galactosidase A inhibitors, such as 1-deoxygalactonojirimycin, at sub-inhibitory concentrations rescued the enzyme activity [46]. Since then, PCs have been identified for several other LSDs. For instance, the identified PC for GM1 gangliosidosis was able to cross the BBB without detrimentally impacting its integrity [52]. Research has continued to identify new PCs and several criteria have been put forward to assure success, including reversible binding to target enzymes, small size, cell permeable, high hydrophobicity (to increase the chances of crossing the BBB and diffusing through refractory tissues), long-term safety, increased stability under ER conditions, and reduced side effects [48]. To find PCs, large chemical libraries of natural or synthesized molecules are either virtually screened using bioinformatics tools or tested in the laboratory [53]. Also, a drug repurposing approach may accelerate the discovery and translation of PC and other small molecules, with a reduced therapy development cost, which can ultimately allow more LSD patients to benefit from a specific therapy [54].

## Hematopoietic stem cell transplantation

HSCT consists of the administration of healthy hematopoietic stem cells to patients, which may be derived from a healthy donor and different sources, including bone marrow, peripheral blood, or umbilical cord blood [55]. The rationale for this treatment is based on the physiological observation that a percentage of the lysosomal enzyme is exported to the extracellular milieu, and subsequently it may be internalized by other cells through the M6PR [56]. HSCT has been proposed for several LSDs since the repopulation of the patient's bone marrow with HSC from a healthy donor may provide a life-long source of unaffected hematopoietic lineage cells expressing normal levels of the deficient enzyme [1]. In vivo, such cell populations include the circulating white blood cells and monocyte-derived macrophages residing within several tissues, including the CNS and bone [57, 58]. HSTC has shown to increase life expectancy and improve or stabilize some clinical manifestations, and in some cases, it is considered as the first choice of treatment, like in the case of MPS I [59]. In fact, for MPS I clinical results suggest that HSCT could work better if the ERT is administered before the transplant [60]. Nevertheless, the impact of HSCT on CNS and skeletal manifestations is still a matter of concern. One major drawback of HSCT is the difficulty to identify a human leukocyte antigens (HLA)-matched donor, summed up to the chance of transplant failure and the high morbidity and mortality rates associated with the conditioning and immunosuppressive regimens, and the potential development of graft-versus-host disease [56, 57].

Although the HSCT can be carried out using progenitors from healthy individuals to heterologous transplants, the use of stem cells obtained from patients with LSDs and genetically modified in the laboratory (*ex vivo* gene therapy) is another explored alternative [61]. This strategy has been tested for GM2 gangliosidosis [62], Fabry [63], and MPS I [64, 65] with promising results such as the improvement of skeletal and neurological abnormalities [66]. With this approach an autologous transplant is possible, which helps avoiding typical limitations of the HSCT like the graft-versus-host disease [67].

## Gene therapy

Gene therapy represents an opportunity to cure the disease by correcting the gene defect by delivering a wild-type copy of the mutant gene via viral or non-viral derived vectors [68]. Viral vectors are commonly derived from a γ-retrovirus, lentivirus, adenovirus, or adeno-associated virus (AAV) [68, 69]. In contrast, non-viral vectors include naked DNA combined with methods that enhance the cell delivery (i.e., ultrasound, electroporation, laser, and gene gun), nanoparticles, and bactofection [70, 71]. Given the monogenic nature of LSDs, gene therapy is shown as the greatest promise to correct these single-gene mutations via direct or cross-correction [72]. Gene therapy effectiveness and gene replacement strategies for LSDs have been extensively demonstrated in an animal model (reviewed [6, 72]). The emergence of genome editing tools, such as CRISPR/Cas9 and zinc finger nuclease (ZFN) has enabled, specific gene targeting, and rewriting. Preclinical studies via these genome editing tools have shown potential for the treatment of LSDs [73–76] and represented a step further towards the consolidation of gene therapy. Preclinical success with AAV-mediated ZFN gene therapy led to phase I/II clinical trial for MPS types I and II [77, 78]. Although these studies evidenced the technique's safety, a number of clinical manifestations [79] have confirmed the importance of improving these gene therapy strategies. We focus here on new treatment alternatives for the LSDs in which genome editing technologies are the mainstay.

**Genome editing** Genome editing uses the intracellular double-strand break (DSB) repair machinery to generate *knockout* or *knockin* through non-homologous end joining (NHEJ) or homology direct repair (HDR), respectively [80]. These mechanisms largely depend on the cell cycle phase. For instance, although the DNA-DSB repair by NHEJ can occur through all the cell cycle, it is most active during G0/G1 and G2 phases, while HDR predominates during S and G2 phases [81]. NHEJ, with the help of factors such as Ku70/80, DNA-

PKcs, Artemis, and DNA ligase IV induces a DNA-DSB repair in a sequence-independent way, thereby making this an error-prone mechanism, since it can induce nucleotide deletions or insertions [82, 83]. In contrast, HDR is considered an error-free mechanism, since during S and G2 phases is present a sister chromatid that can be used as a template for repair through the strand invasion mediated by the recombinase RAD51 [84]. Furthermore, it has been found that BRCA2 plays a vital role in the formation of RAD51-mediated nucleofilaments, strand exchange in mammalian cells, and stabilization of ATPase activity of RAD51 [81]. In this sense, genome editing via HDR must be carried out with caution in patients with BRCA2 mutations [81, 84]. Although NHEJ and HDR can be used to repair DNA-DSBs, gene therapy of LSDs is a more interesting approach to induce HDR, since it will allow the integration of a wild-type copy of the mutated gene. For this purpose, the therapeutic DNA must be flanked by homologous arms to the desired integration site and to stimulate the homologous recombination in the cell. This approach can be performed through different genomic edition tools such as ZFN, transcription activator-like effector nucleases (TALENs), or CRISPR/Cas9, which will be described below.

**Zinc finger nucleases** ZNF are synthetic chimeric proteins composed of zinc finger proteins and FokI nucleases [85]. Structurally, these proteins are composed of modules of 30 amino acids which are folded into a finger-like structure through the interactions between a zinc atom and two conserved cysteine ($Cys_2$) and histidine ($His_2$) residues [80]. Each module can recruit a single zinc atom to fold the peptide in a ββα conformation that can recognize 3 to 4 DNA nucleotides. Due to this property, a tandem array of zinc fingers can be designed and fussed to FokI nuclease such that site-directed DSB is induced within the host genome (Fig. 2a) [86].

ZFNs have been used in Gaucher [87], Fabry [87], MPS I [88], MPS II [89], and mucolipidosis IV [90]. This has been achieved with the albumin intron 1 as the locus insert the cDNA of α-galactosidase A (Fabry), α-L-iduronidase (Hurler), iduronate-2-sulfatase (Hunter), or acid-β-glucosidase (Gaucher). In this case, ZFN was delivered with the aid of AAV8 vectors in wild-type mice to obtain supraphysiological levels of these enzymes in liver [87]. Surprisingly, only β-glucosidase levels showed a significant increase in plasma. Although no therapeutic effects were assessed, this work provides evidence that albumin intron could be a safe harbor for the insertion of multiple transgenes [87]. Recent studies, supported by Sangamo Therapeutics, reported the insertion of α-L-iduronidase (IDUA) cDNA within the albumin intron 1 locus via ZFN in a MPS I mouse model with a single tail-vein injection of $1.5 \times 10^{12}$ vg/mouse [76]. Following the same strategy, the iduronate 2-sulfatase gene was integrated into an MPS II mouse model using three different AAV doses, namely low ($2.5 \times 10^{11}$ vg/mouse), medium ($5 \times 10^{11}$ vg/mouse) or high ($1.5 \times 10^{12}$ vg/mouse) [75]. Compared to untreated mice, IDUA activity was increased in the spleen, heart, lung, and muscle, while dermatan sulfate, but not heparan, was reduced in the brain of MPS I mouse [76]. Similarly, in treated MPS II mice, the IDS activity increased up to 207-fold compared to untreated mice, while GAGs decreased to wild-type levels in several organs such as the liver, spleen, kidney, lungs, heart, and muscle [75]. In the brain, GAGs remained like untreated MPS II mice. Moreover, high doses of ZNF-donor formulation induced higher enzyme activity in the brain, ~ 1.5% of wild-type, and improved cognitive capacity. These results suggest that ZFN-mediated gene correction, using album intron 1 as a safe harbor, is an alternative to overcome the main limitation of therapies for neurodegenerative LSDs since the lysosomal enzyme could be synthesized by the liver, released to circulation, and be captured by other organs, like the brain [75, 76].

Previous findings allowed starting a phase 1/2 multicenter clinical trials for MPS I (ClinicalTrials.gov Identifier: NCT02702115) and MPS II (ClinicalTrials.gov Identifier: NCT03041324) of a single administration of a ZFN delivered via AAV vectors. In the case of MPS I, three adult patients were enrolled in a phase 1/2 clinical trial to evaluate the safety and tolerability of ascending doses of the ZFN therapy SB-318. No adverse effects related to SB-318 were reported. Although IDUA activity was increased in leukocytes 22 weeks post-treatment, no improvement was observed in plasma IDUA activity or urinary GAGs [88]. Similarly, the MPS II phase 1/2 clinical trial was conducted in eight patients to evaluate the safety, tolerability, and effect on leukocyte and plasma IDS activity of ascending doses of the ZFN therapy SB-913. The results indicated that the therapy was generally well-tolerated with no reports of serious adverse effects. Nevertheless, plasma activity was only increased in one patient, which returned to baseline levels after the development of mild transaminitis. The reduction of urinary GAGs was not observed for 24 weeks post-treatment [89]. Overall, these results show that genome editing by ZFN is safe, but the therapeutic efficacy needs to be improved.

Although no recent therapeutics approaches have been published for ZFN in the treatment of LSDs, ZFN has also been explored in the development of disease animal models. For instance, ZFN has been used in zebrafish to *knockout* the *mcoln1a* gene, which encodes a lysosomal cation channel that is deficient in mucolipidosis type IV (ML IV). The obtained model reproduced well the typical clinical findings observed in ML IV patients [90].

**Transcription activator-like effector nucleases** Transcription activator-like effector (TALE) protein is derived from *Xanthomonas* spp., a plant pathogenic bacteria [80]. This protein has a DNA-binding domain consisting of

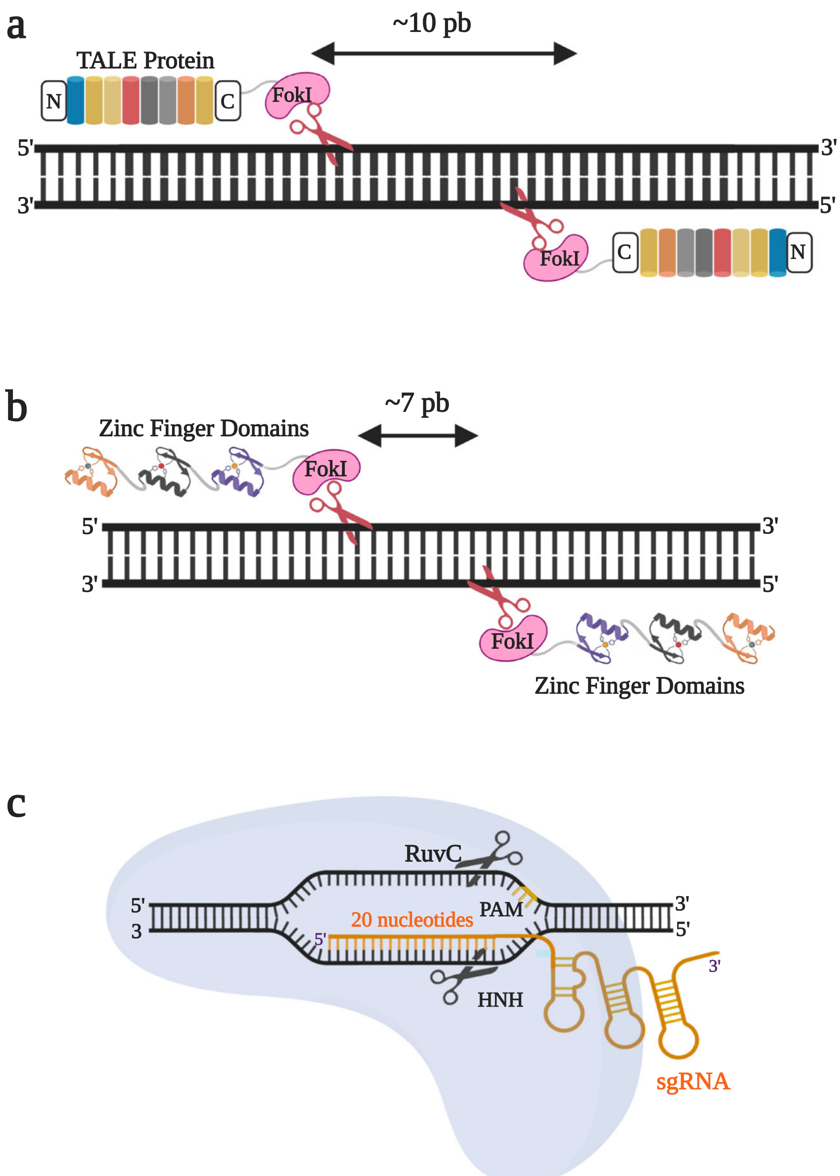


**Fig. 2** Genome editing tools. **a** Zinc finger domains consists of 30 amino acids with ββα conformation which has a classical structure with Cys2, His2 zinc finger motif. Engineering design is possible by the change of amino acids in the zinc-finger motif to recognize and interact with specific DNA sequences. ZF are fussed to nucleases like FokI. **b** Transcription activator-like effectors (TALE) are a tandem of highly conserved repeats of 34 amino acids. Standard strategy needs dimerization of catalytic domain of Type IIS nucleases like FokI to induce a DSB in target DNA, and higher efficiency is reached when introducing a spacing of at least 10 bp between both nucleases. **c** Clustered regularly interspaced short palindromic repetitions (CRISPR) as genome editing tools uses a single guide RNA (sgRNA) designed to find a complementary sequence in DNA target. sgRNA is found associated with Cas9, an endonuclease that scan the host for PAM sequences (usually NGG, where N is anyone nucleotide). Wild-type Cas9 has two catalytic domains (RuvC and HNH) that cut the sense and antisense strand of DNA and generate a DSB. This figure was created with BioRender.com

highly conserved tandem repeats of 34 amino acids each one. Additionally, it contains two variable residues in the positions 12 and 13, which confer a DNA-specific base recognition and that can be easily and rapidly altered. This finding allowed the design of a genome editing system composed of a TALE fused to nucleases (TALENs), such as FokI or TevI. These nucleases are directed by two amino acids (TALE code) to induce specific DSBs in a desired region of interest. High efficiency is reached when spacers of 10 bp are introduced between both nucleases (Fig. 2b) [91]. TALENs have been used for gene editing in several models such as yeast [92], zebrafish [93], frog [94], and human cells [95, 96]. In the case of LSDs, TALENs have been used for the development of cellular and animal model. For instance, human-induced pluripotent stem cells and zebrafish have been used to develop Gaucher disease models [97, 98]. To date, this tool has not been used in preclinical or clinical studies of LSDs.

**Clustered regularly interspaced short palindromic repetitions** CRISPR could be thought as an adaptive immune system in bacteria against phages such that when a virus re-infection takes place, RNA fragments (crRNA) and a bacterial scaffold RNA (tracrRNA) are transcribed and complexed with Cas9 [99]. Cas9 is an enzyme with two endonucleases domains, HNH and RuvC, which are capable of cleaving the complementary and non-complementary strands, respectively [80]. Mature crRNA alone cannot direct Cas9-catalyzed DNA cleavage and needs the presence of an RNA structure (single-guide RNA (sgRNA)) and a protospacer adjacent motif (PAM) juxtaposed to the complementary region in the target DNA [100]. These findings triggered a revolution in genome editing strategies where specific DNA is targeted by customized sgRNA sequences (~ 20 nucleotides) fused to tracrRNA to direct the (Fig. 2c). Modifications in HNH and RuvC domains have allowed increasing the specificity of the system [101]. This genome editing tool requires the DNA repair process, which requires a careful assessment of the state of the cell chromatin for *in vitro* and in vivo approaches. Also, it is important to note that the presence of nucleosomes prevents CRISPR/Cas9 to access the genome and consequently, a significant reduction in its activity might be observed [102]. Despite the advances in the use of TALEN and ZFN for LSDs, CRISPR/Cas9 continues to be the leading technology for genome editing LSDs as evidenced by, 28 publications between 2016 and 2020. Twenty of these publications are related to the development of disease models (Table 1), while the remaining ones are associated with the development of therapeutic approaches.

**CRISPR/Cas9 and MPS** Therapeutic advances with CRISPR/Cas9 for LSDs have been focused on MPS I, including a preliminary validation for ex vivo genome editing [121]. Recently, a cassette containing the IDUA gene flanked by homologous arms for the CCR5 gene was inserted into CD34$^{+}$ hematopoietic stem and progenitor cells (HSPC) via CRISPR/Cas9 [122]. As a result, the enzymatic activity of IDUA in HSPC was increased 600-fold, while in HSPC-derived macrophages it reached an increase of 75-fold. Furthermore, IDUA$^{-/-}$ mice transplanted with the edited cells reported an increase IDUA activity of about 6.8%, 11.3%, 50.1%, and 167.5% in the brain, serum, liver, and spleen, respectively, in addition to a reduction of excreted GAGs of about ~ 65% when compared to the sham group after 18 weeks post-transplantation [122]. Although this strategy may be a promising alternative for the treatment of MPS I, transplantation in humans has several limitations such as the

**Table 1** Use of CRISPR/Cas9 in development of disease models for LSDs

| Type | Disease | Affected gene | Affected protein | Model | Reference |
|---|---|---|---|---|---|
| Sphingolipidosis | Fabry | *GLA* | α-Galactosidase A | Podocyte | [103] |
| | | | | EA.hy926 cells | [104, 105] |
| | | | | hESC | [106] |
| | | | | HEK-293 cells | [107] |
| | Gaucher | *GBA* | β-Glucocerebrosidase | HEK293 cells | [108] |
| | | | | THP-1 and Glioblastoma cells | [109] |
| | | | | Zebrafish | [98] |
| | GM1 gangliosidosis | *GLB1* | β-Galactosidase | iPSC | [110] |
| | | | | C57BL/6 | [111] |
| | Niemann-Pick-type C | *NPC1/NPC2* | Intracellular cholesterol transporter 1/2 | HAP1 cells | [112] |
| | | | | Zebrafish | [113, 114] |
| | | | | HeLa | [115] |
| | Sandhoff | *HEXB* | β-Hexosaminidase A and B | iPSC | [116] |
| Other lipidosis | Wolman/CESD | *LIPA* | Acid lipase | iPSC | [117] |
| Mucopolysaccharidosis | Morquio IVB | *GLB1* | β-Galactosidase | C57BL/6 | [111] |
| Glycogenosis | Type II (Pompe) | *GAA* | α-Glucosidase | iPSC | [118] |
| Neuronal ceroid lipofuscinosis | CLN1 | *PPT1* | Palmitoyl-protein thioesterase 1 | Sheep | [119] |
| | CLN3 | *CLN3* | CLN3 protein | iPSC-derived organoids | [120] |

*CESD*, cholesteryl ester storage disease; *hESC*, Human embryonic stem cells; *iPSC*, Induced pluripotent stem cells hyperacute rejection associated with high mortality, which even with an HLA match donor might lead to persistent osteoarticular residual disease [123]. Also, the use of CCR5 locus for DNA donor integration could have several ethical implications, because CCR5 deletion (CCR5$^{\Delta 32}$) has been implicated in recovery after stroke in mice [124]. CCR5 disruption could also increase cognitive ability in the edited individuals. This field is still restricted by the international community [125] and will be discussed later.

Alternatively, the albumin locus has been assessed as a harbor for the insertion of IDUA cDNA via AAV8-derived vectors carrying a CRISPR/Cas9 system [126]. Neonatal MPS I mice treated with an intravenous administration of these modified AAV8 vectors achieved a significant decrease in the level of GAGs in different tissues. Furthermore, a considerable increment in IDUA activity was observed in several organs including the brain that along with the learning test that gets back to normal shows the positive impact of this therapy on the CNS [126]. However, this treatment caused a marked decrease in albumin levels in serum, which returned to normal levels 10 months post-treatment. Noteworthy, neither tumorigenesis nor *off-target* effects were found in treated animals [127]. These findings suggest that, as reported for ZFN, albumin locus could be a safe harbor for CRISPR/Cas9, which suggest that the liver could be exploited as a factory to obtain the significant amounts of the therapeutic enzyme needed to reach the CNS. However, future use of CRISPR/Cas9 in humans must consider the impact of transitory hypoalbuminemia observed in pre-clinical models [87].

In MPS I, c.1205G>A (p.Trp402*) is a frequent mutation in South American patients that introduces a premature stop codon in the position 402 of IDUA enzyme by a single base substitution (T**G**G to T**A**G) [73]. The use of CRISPR/Cas9 and a single-strand wild-type donor sequence homologous to the p.Trp402* region in human fibroblasts allowed recovery of ~ 4% of enzymatic activity in fibroblasts and extracellular medium. This was complemented by a significant decrease of lysosomal mass [73]. In this study, nano-emulsions were used as a non-viral delivery system. Later, this approach was tested in newborn MPS I C57BL/6 mice, using a single nano-emulsion shot in the superficial temporal vein to target the Rosa26 locus as a safe harbor [74]. IDUA activity in serum from treated mice was 6% of the wild-type group and the enzyme was distributed to the lungs, heart, liver, kidneys, and spleen, which seemed to contribute to reducing the overall GAGs levels. However, in the brain, no correction of enzyme deficiency was observed [74]. To overcome this pitfall, the nasal administration of the nano-emulsion was proposed obtaining IDUA enzyme activities 2-fold higher in the brain of treated MPS I mice compared to untreated mice [128]. Similar findings were recently reported with the delivery of these nano-emulsions via hydrodynamic injection in neonatal MPS I mice [129]. Reduction in GAGs levels in the heart, kidney, lungs, and spleen was reported, as well as the improvement of physiological parameters such as lung resistance and behavior test [129]. These results suggest that the brain delivery of CRISPR/Cas9 aided by nano-emulsions upon nasal administration could overcome the biodistribution limitations of typical therapeutics alternatives for MPS I, such as ERTs, which generally fail to reach the brain, an essential organ due to the neurological deficiencies seen in MPS I patients.

**CRISPR/Cas9 and GM2 gangliosidosis** GM2 gangliosidosis is a group of three neurodegenerative diseases caused by a deficiency of β-hexosaminidases (Tay-Sachs or Sandhoff disease) or the GM2 activator protein (AB variant disease), leading to the lysosomal accumulation of GM2 ganglioside [130]. A recent study using CRISPR/Cas9 and albumin locus as a safe harbor evaluated the therapeutic effect of a chimeric protein termed HexM, which contains the α-subunit active site and the stable β-subunit interface to degrade the GM2 ganglioside [127]. Using a hydrodynamic injection to deliver the HexM cDNA into a Sandhoff disease animal model, the treated group showed a ~ 7-fold higher enzymatic activity than the untreated control group in several tissues such as the liver, heart, and spleen. Nonetheless, both a reduction in GM2 ganglioside levels was observed in the brain or an improvement in the pole and fear test, which indicates this genome editing tool requires further improvement [127]. Importantly, this study adds evidence that supports the use of albumin locus as a safe harbor for gene integration to correct the enzyme deficiency in several tissues.

Despite the beneficial results that CRISPR/Cas9 may offer for the treatment of LSDs, few studies have evaluated this genome editing tool for the treatment of LSDs as well as for other inborn errors of metabolism [66]. Furthermore, there is scarce information about more potential genes that can act as safe harbors for gene integration, as well as the assessment of target-specific delivery carriers. In this regard, our laboratory is currently evaluating the therapeutic use of CRISPR/nCas9 (a variant of Cas9 wild-type, which has a mutation in RuvC domain) in fibroblasts from patients with MPS IVA, phenylketonuria, Sandhoff, and Tay-Sachs, using magnetoliposomes as non-viral delivery vectors [131].

## Advantages and challenges of genome editing tools

The development of custom-designed nucleases, including ZFNs and TALENs, to perform genetic engineering and more recently, CRISPR/Cas9, have revolutionized the gene therapy. However, several challenges have been described for each one. Here, we discuss the significant advantages and challenges of each one, which are summarized in Table 2.

**Table 2** Advantages and challenges of major genome editing tools

| Genome editing tool | Advantage | Challenges | References |
|---|---|---|---|
| ZFNs | High specificity<br>Compatible with viral vectors<br>Any genome sequence can be targeted | Time-consuming<br>Low efficiency<br>Target one site each time<br>Complex and expensive design of endonucleases<br>High cytotoxicity | [85, 132] |
| TALENs | High specificity<br>Low *off-target* effect<br>Any genome sequence can be targeted | Time-consuming<br>Large size<br>Low efficiency<br>Genotoxic effects<br>Target one site each time<br>Sensitive to target DNA methylation | [133–135] |
| CRISPR/Cas9 | High specificity<br>High efficiency<br>Easier design<br>Cost-effectiveness<br>Multiple sites can be edited at the same time | *Off-target* effects<br>Mosaicism<br>Low stability of sgRNA<br>Gene target requires a PAM | [136–139] |

ZFNs are a first generation of gene editing techniques based on a modular assembly strategy. However, several drawbacks have been observed following their use both in vitro and in vivo. For instance, the self-interaction of its amino acid causes a reduction in the efficiency of gene targeting [140]. Thus, there is a "context-dependent effect" for the interaction of the side chain of aspartate to favor the recognition of G or T as the first base pair (bp) of the previous finger's subsite, when positioned as the second residue of the α-helix [141]. To overcome these context-dependent effects, several approaches have been developed such as the sequential selection method, where each new finger is selected according to its neighbor [142] or the use of a bacterial two-hybrid system, in which there is an enrichment of fingers on out-of-context 3-bp subsites, followed by a final selection of enriched fingers in the context of the full intended DNA target sequence [143]. Although the obtained zinc fingers have a superior affinity and specificity than those produced by the classical modular assembly, they are more laborious to obtain. Besides, the off-targets have shown to be dependent on the concentration of ZFN in the system [144] and might result in important cytotoxic effects [145]. Then, designing a suitable ZFN for a specific target gene is laborious, and not all genes in the genome can be edited by this tool [146]. Compared to traditional homologous recombination, ZFN technology has a gene targeting efficiency that ranges between 10 and 30% [132].

TALENs, compared to the ZFN approach, has higher targeting efficiency (between 20 and 60%) [94], higher specificity with little off-target effect, and lower nuclease-associated cytotoxicity [133]. Nevertheless, TALENs have shown limited efficacy targeting specific loci in the human genome [134]. Then, a second generation of TALENs, with improved human genome editing efficacy, has been developed via direct evolution in which the C-terminal segment and plays the leading role in the activity improvement [135]. Despite the advantages presented by the second generation of TALENs, there is still chances for increased DNA damage, which addressed by reducing the TALEN concentration [135]. Compared to ZFN, TALENs present a large molecular size, which limit operationality at a molecular level [133, 147]. Despite the ease to design TALENs for a specific target DNA sequence than ZFNs, their assembly is time-consuming and requires extensive sequencing to ensure the correct design and DNA sequence targeting, which ultimately leads to extra cost during development [146].

Among the genome editing tools, CRISPR/Cas9 has the highest targeting efficiency ranging from 50 to 80% [136]. Targeting site design, vector construction, and operation of CRISPR/Cas9 is easier than for ZFNs and TALENs. Compared to ZFNs and TALENs, CRISPR/Cas9 can target any genomic sequence by changing the 20 bp protospacer of the guide RNA [148]. In addition, simultaneous targeting of multiple sites can be easily addressed with the CRISPR/Cas9 technology [137]. In the case of CRISPR/Cas9, DNA recognition appears to lead to a reduction in off-target effects, which is mainly due to the base pairing targeting mechanism that relies on interaction between the target DNA and RNA [149]. Despite the reduction in the presence off-targets, this remains as a major issue for the further translation of CRISPR-Cas9 systems as potential therapies. Recent studies have, in fact, shown that the *S. pyogenes* CRISPR/Cas9 (SpCas9) system can tolerate up to three mismatches without impairing the ability to cleave DNA in a sequence-dependent manner [138, 150]. To overcome these drawbacks, several approaches have been proposed such as the rational design of sgRNAs based on bioinformatics predictions [150] and use of alternative Cas9 nucleases such as the mutated Cas9 variants [139, 151] or the one isolated from *Neisseria meningitidis* that binds a 24 bp protospacer sequence on its target DNA. This results in a superior specificity than that of the SpCas9 [152]. Potential immunogenicity of Cas proteins following its delivery and tumorigenicity are major concerns for the use of CRISPR/Cas9 in clinic applications [66]. This has been documented in animal models, specifically in mice [153, 154]. However, the implications of this immunogenicity for therapeutic applications remain unclear.

## Bioethical aspects of genome editing

Genome editing methodologies have gained considerable attention and even evolved in the last years as potent tools, to rewrite the genome in vitro and in vivo model organisms. Moreover, such tools have implemented to tweak gene expression programs, and drive drug development and disease etiology research [155]. The important potential of such editing technologies has brought expectations for the efficient healing of several genetic diseases. In addition to the technical challenges previously described, ethical implications related to these technologies began to emerge ever since the first promising results were observed [156]. One major concern associated with the use of this technology is the potential transmission of gene editions to germline cells as they can cause unexpected impacts to future generations [157].

Although researchers have manipulated germlines for several years, the real controversy occurs when these methodologies are evaluated in humans, and when genome editing goes beyond the modification of somatic cells, involving heritable alterations [155]. To avoid the use of genome editing technologies in human embryos and germline cells, policies about the right use of these tools have been put forward, and extensively and interdisciplinary discussed by researchers, lawyers, clinicians, and ethicists. Nevertheless, two controversial research articles about the use of CRISPR/Cas9 in human embryos discussed the edition of the β-globin gene, which is involved in β-thalassemia, which introduces a naturally occurring CCR5$^{\Delta 32}$ allele in humans zygotes [158, 159]. Both research studies obtained low HDR efficiency, embryos genetic mosaicism, and several off-target mutations. Noteworthy, ethical committees approved both studies.

Furthermore, to avoid additional ethical concerns and criticisms, tripronuclear zygotes were included in the experimental set, which failed to develop normally and were ultimately discarded during the in vitro fertilization process [157]. The failure of the experiments in human embryos spurred significant controversy and showed no evidence if the safeness of the therapies due to the large number of off-targets generated along the process [157]. For these reasons and the generated off-target heterogeneity, it is necessary to propose new strategies to increase the specificity of the involved nucleases as well as precise transcriptional control and integration of new sequences [160]. These unsolved inconsistencies remain have limited the implementation of genome editing technologies for therapeutics purposes in germline cells yet. Besides, these tools potentially promote issue in terms of social values and inequality considering that genome editing technologies might be initially accessible to high-income countries and individuals [161].

Using gene editing tools in nonviable human embryos has raised serious discussions on ethical implications that demand stringent regulations from authorities [156]. Several years ago, the National Academy of Sciences (NAS) and the National Academy of Medicine (NAM), in conjunction with the Royal Academy of Sciences and the Chinese Academy of Sciences, assembled a "Committee on Human Gene Editing: Scientific, Medical, and Ethical Considerations" to answer some questions related to the application of genome editing tools in humans [156]. Together, NAS and NAM made the following recommendations for the use of genome editing tools: (i) to avoid the mutation of genes associated with common health problems, for which little or no evidence of additional adverse effects that this entails is available; (ii) to avoid editing genes that could cause a predisposition to a specific disease; (iii) to implement a rigorous control at each step from in vitro assays to clinical trials, ensuring the safety of research participants; (iv) to avoid clinical trials without enough preliminary data about potential procedure risks over health benefits; (v) to guarantee multigenerational follow-up, including a health evaluation and social benefits and risks; and (vi) to implement reliable oversight mechanisms to prevent the non-therapeutic use of the technology [125].

## Conclusions and perspectives

New therapeutic strategies have been developed to treat LSDs. Currently, several treatment options are available or under development. ERT is the main treatment option for LSD patients, for which new enzymes have been approved during the last few years. These developments have led to eleven treated disorders. However, current ERT approaches present several limitations. This has spurred the search for new targeting strategies, especially with a focused on hard-to-treat tissues and the engineering of enzymes with improved activity and half-life. Furthermore, recent studies have aimed at reducing the immunological issues and to improving the cost-benefit relation. Although small molecules (i.e., SRT and PCs) have offered an opportunity to develop novel treatment strategies, based on the potential of oral delivery and BBB crossing, only a reduced number of them have reached in vivo assessment. Consequently, considerable effort will be needed to assure the evaluation of these molecules at the clinical trials levels.

The recent approval of gene therapy-based drugs by the FDA and EMA has attracted the attention of different stakeholders again to this field. Moreover, the development of genome edition has offered new tools to improve the safety and efficacy of gene therapy. Although animal studies have shown the potential of genome editing for the treatment of LSDs, their clinical impact stills remain elusive. Among the different genome editing tools, CRISPR/Cas9 has emerged as one of the most versatile and easy tools to be adapted and further implemented. Current reports have shown that CRISPR/Cas9 can increase the enzymatic activity to therapeutic levels in animal models, thereby leading to a significant reduction in the levels of stored substrates.

Nevertheless, like other therapies for LSDs, the treatment of brain and bone pathologies remains as an open challenge. Besides, as for traditional gene therapy approaches, perhaps the main challenge of genome editing technologies is the low delivery efficiency of nucleic acids. Currently, delivery strategies of genome editing tools rely on the use of viral vectors, which exhibit several limitations including cargo size, tissue targeting, random genome integration versus episomal expression, and production costs. Although some studies have shown the potential of non-viral vectors as delivery vehicles for genome editing tools, it still challenging to engineer systems capable of carrying two or more expression vectors as they are needed for most therapies. However, the ease adaptation and manipulation of genome editing tools may offer an alternative to accelerate the development of treatment alternatives for LSDs, mainly for those where no specific treatment is currently available.

**Funding information** CJAD, AJEM, LHR, and JCC are supported by the Ministry of Science, Technology and Innovation, Colombia (Grant ID 120380763212—PPTA # 8352). CJAD and AJEM are supported by Pontificia Universidad Javeriana (PPTA # 8275). AFL received a doctoral scholarship from Pontificia Universidad Javeriana. OFS received a postdoctoral fellowship from the Ministry of Science, Technology and Innovation, Colombia (#811-2018).

## Compliance with ethical standards

**Conflict of interest** The authors declare that they have no conflict of interest.

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